Cover Letter: JMLR Machine Learning Open Source Software

kodama-cpp: Cross-Validated Accuracy Maximization on CPU, CUDA, and Apple Metal

Dear Editors of the Journal of Machine Learning Research,

We submit kodama-cpp for consideration in the Machine Learning Open Source Software track. The software is a standalone C++17 implementation of KODAMA with explicit CPU, NVIDIA CUDA, and Apple Metal backends and thin R and Python wrappers.

Submission identity

Open-source license: MIT project code with compatible retained third-party terms; see PROVENANCE.md and THIRD_PARTY_NOTICES.md

Project URL: https://github.com/tkcaccia/kodama-cpp

R wrapper URL: https://github.com/tkcaccia/kodama-r

Python binding source: https://github.com/tkcaccia/kodama-cpp/tree/main/split-repos/kodama-python (standalone repository publication pending)

Software version: 0.1.0 release candidate

Reviewed baseline: a32f71844fb0860f8faec0169c2850f800a68da4

Reviewed R-wrapper baseline: 4d5506bde8918162dae0ad1d19c4f6f7a4df64ba

Final release commit: AUTHOR ACTION REQUIRED: insert the full v0.1.0 tagged commit

Archived source: AUTHOR ACTION REQUIRED: deposit the tagged source archive, insert its URL and checksum, and add a DOI if assigned

Significance and software contribution

The submission preserves KODAMA's cross-validated label-predictability signal while making four contributions: a standalone C++17 core for thin R/Python wrappers; native and observable CPU/CUDA/Metal execution; guided label evolution based on out-of-fold predictions and class transitions; and an exact-quota landmark-projection strategy that generalizes the landmark idea introduced in the KODAMA bioRxiv work. The repository contains installation instructions, API examples, explicit pseudocode, tests, benchmark drivers, a public API snapshot, and a machine-checkable licensing/provenance audit.

Prior publications and delta

The KODAMA method was published by Cacciatore, Luchinat, and Tenori in PNAS (2014), and the historical R package was described by Cacciatore et al. in Bioinformatics (2017). Prior publication of the method is disclosed. The landmark-projection principle was subsequently introduced in the 2025 KODAMA bioRxiv preprint. This submission concerns its general formalization and the new standalone software: a C++17 core independent of R, native CUDA and Metal backends, guided search mechanics, exact-quota landmark selection, float32 label-aware SIMPLS/LDA, package-owned neighbor search, strict backend metadata, and separately maintained R/Python bindings. KODAMA R 2.4.1/2.4 is the sole previous-version comparator; later KODAMA releases are excluded from performance, quality, and methodological comparisons. Quantitative predecessor comparisons are restricted to classifier-matched KNN because historical PLS-DA is not equivalent to current PLS-LDA. The detailed delta is recorded in docs/previous-version-delta.md. The prior papers and preprint will be supplied as supplementary material.

Evidence of use and openness

KODAMA has an established public predecessor: a peer-reviewed method paper, a peer-reviewed R-package paper, a maintained public R package, documentation, and public source history. The CRANlogs public API reports 48,480 KODAMA downloads from 2017-01-01 through 2026-07-26, including 4,526 during 2025-07-27--2026-07-26 (https://cranlogs.r-pkg.org/downloads/total/2017-01-01:2026-07-26/KODAMA and https://cranlogs.r-pkg.org/downloads/total/2025-07-27:2026-07-26/KODAMA). These counts concern the predecessor R package, not the new implementation. kodama-cpp itself is a new repository and, on 2026-07-27, had zero GitHub stars, forks, releases, and public issues; these figures are disclosed rather than presented as adoption. AUTHOR ACTION REQUIRED: refresh the dated counts and add any independently verifiable user or downstream-package evidence available immediately before submission. The repository provides an issue tracker, contribution guide, changelog, API-stability policy, reproducible tests, and release checklist.

Required author declarations

- Coauthor consent: AUTHOR ACTION REQUIRED. Confirm that all seven authors know of and consent to this submission.
- Funding supporting this work during the previous 36 months: AUTHOR ACTION REQUIRED.
- Competing interests and recent collaborations with JMLR action editors: AUTHOR ACTION REQUIRED.
- Suggested action editors: AUTHOR ACTION REQUIRED. Supply 3-5 conflict-free JMLR action editors with brief relevance statements.
- Suggested reviewers: AUTHOR ACTION REQUIRED. Supply 3-5 conflict-free reviewers with brief relevance statements.

Keywords

KODAMA; cross-validation; unsupervised learning; heterogeneous computing; manifold learning

Sincerely,

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